



Problem Solving With C - UE25CS151B

Pointers

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1. What is a Pointer?
2. Declaration and Initialization
3. Arithmetic operations on Pointer
4. Array Traversal using pointers
5. Array and Pointer

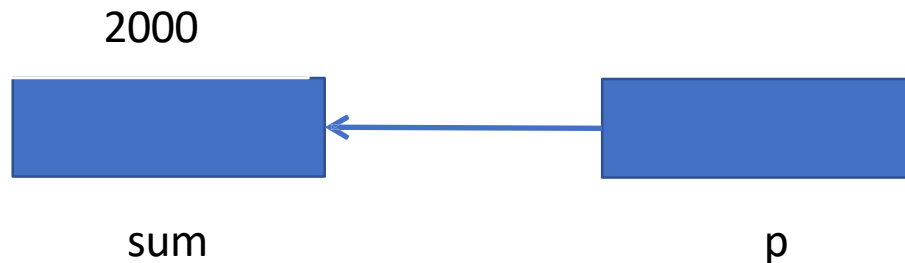
What is a Pointer?

- A variable which contains the address. This address is the location of another object in the memory
- Used to access and manipulate data stored in memory.
- Pointer of particular type can point to address of any value in that particular type.
- Size of pointer of any type is same/constant in that system
- Not all pointers actually contain an address
Example: NULL pointer // Value of NULL pointer is 0.
- Pointer can have three kinds of contents in it
 - The address of an object, which can be dereferenced.
 - A NULL pointer
 - Undefined value // If p is a pointer to integer, then – int *p;

Declaration and Initialization

Declaration Syntax: Data-type *name;

- `int *p;`
 - Compiler assumes that any address that it holds points to an integer type.
- `p = ∑`
 - Memory address of sum variable is stored into p.



Example code:

```
int *p; // p can point to anything where integer is
        stored. int* is the type. Not just int.
int a = 100;
p=&a;
printf("a is %d and *p is %d", a,*p);
```



Pointer Arithmetic Operations

1. Add an int to a pointer
2. Subtract an int from a pointer
3. Difference of two pointers when they point to the same array.

Note: Integer is not same as pointer.

- Coding examples

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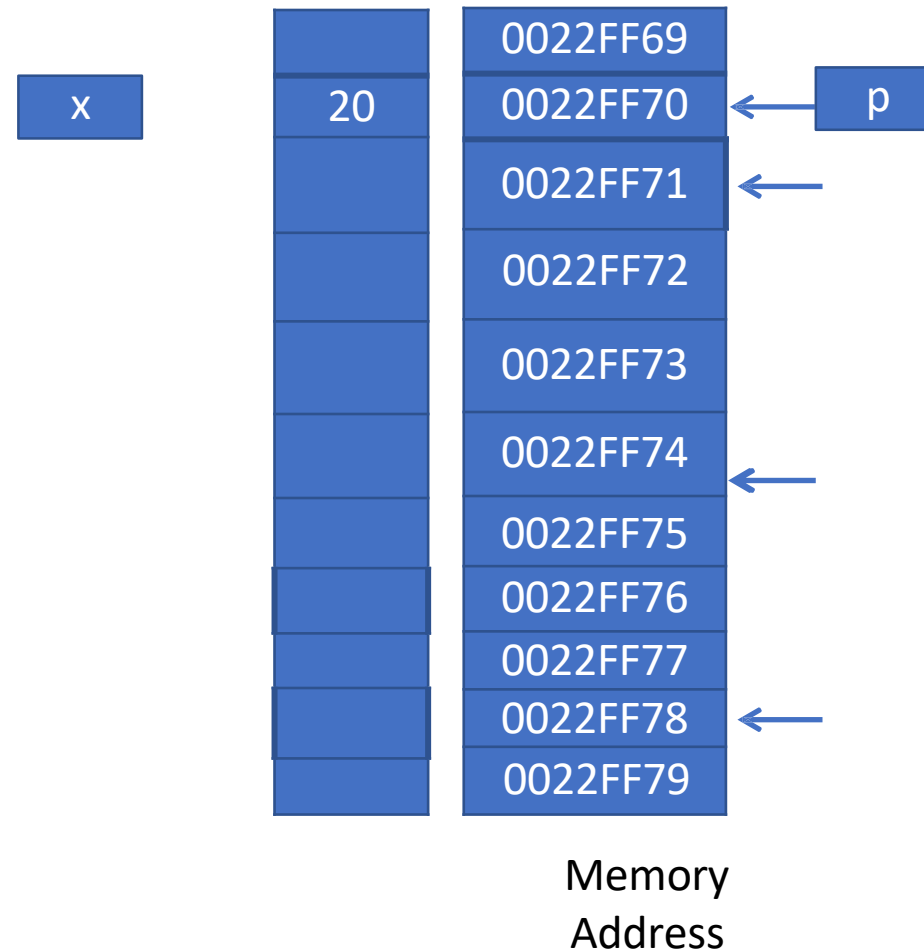
Pointer Arithmetic Operations continued..

Example Code:

```
int *p, x = 20;  
p = &x;  
printf("p    = %p\n", p);  
printf("p+1 = %p\n", (int*)p+1);  
printf("p+1 = %p\n", (char*)p+1);  
printf("p+1 = %p\n", (float*)p+1);  
printf("p+1 = %p\n", (double*)p+1);
```

Sample output:

```
p    = 0022FF70  
p+1 = 0022FF74  
p+1 = 0022FF71  
p+1 = 0022FF74  
p+1 = 0022FF78
```



Array Traversal using Pointers

Consider `int arr[] = {12,44,22,33,55};` `int *p = arr;` `int i;`

Coding examples to demo below points

- Array notation. Index operator can be applied on pointer.
- Pointer notation
- Using `*p++`
- Using `*++p`, Undefined behavior if you try to access outside bound
- Using `(*p)++`
- Using `*p` and then `p++`

Array and Pointer

- An array during compile time is an actual array but degenerates to a constant pointer during run time.
- Size of the array returns the number of bytes occupied by the array. But the size of pointer is always constant in that particular system.

```
int *p1; float *f1 ; char *c1;  
printf("%d %d %d ",sizeof(p1),sizeof(f1),sizeof(c1)); // Same value for all
```
- An array is a constant pointer. It cannot point to anything in the world

Array and Pointer continued..

Example code:

- `int a[] = {22,11,44,5};`
- `int *p = a;`
- `a++;` // Error : a is constant pointer
- `p++;` // Fine
- `p[1] = 222;`
- `a[1] = 222 ;` // Fine
- If variable i is used in loop for the traversal, `a[i]`, `*(a+i)`, `p[i]`, `*(p+i)`, `i[a]`, `i[p]` are all same.

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Array and Pointer continued..

Differences

1. the sizeof operator
 - a. sizeof(array) returns the amount of memory used by all elements in array
 - b. sizeof(pointer) only returns the amount of memory used by the pointer variable itself
2. the & operator
 - a. &array is an alias for &array[0] and returns the address of the first element in array
 - b. &pointer returns the address of pointer
3. string literal initialization of a character array
 - a. char array[] = "abc" sets the first four elements in array to 'a', 'b', 'c', and '\0'
 - b. char *pointer = "abc" sets pointer to the address of the "abc" string (which may be stored in read-only memory and thus unchangeable)
4. Pointer variable can be assigned a value whereas array variable cannot be.
5. Arithmetic operations on pointer variable is allowed. On array, not allowed all.



THANK YOU

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